

Complete Data Cleaning & Balancing Pipeline - Summary

Purpose

This pipeline prepares a perfectly balanced, cleaned dataset for Machine Learning training by:

- Taking your existing imbalanced SMS dataset
- Generating synthetic data to balance it
- Cleaning and preprocessing text
- Creating a ready-to-use ML dataset

Dataset Transformation Summary

Metric	Original Dataset	Final Dataset
Total Messages	Your original count (imbalanced)	100,000 (perfectly balanced)
Spam	Imbalanced (fewer spam)	50,000 (50%)
Ham	Imbalanced (more ham)	50,000 (50%)
Balance	✗ Imbalanced	✓ Perfectly Balanced

Key Features

1. Data Augmentation

- Generates realistic synthetic spam and ham messages
- Uses 35+ spam templates with 12+ keyword categories
- Uses 30+ ham templates with 12+ keyword categories
- Creates messages with realistic patterns

2. Smart Balancing

- **Oversampling:** Creates more samples of minority class
- **Undersampling:** Reduces samples of majority class
- **Target:** Exactly 50% spam, 50% ham

3. Text Cleaning for ML

- ✓ Converts to lowercase
- ✓ Removes special characters (keeps alphanumeric)
- ✓ Removes stopwords (EXCEPT spam indicators)
- ✓ Applies lemmatization
- ✓ Keeps important spam keywords

4. Spam Indicators Preserved

Keeps important words like:

text
free, win, winner, cash, prize, claim, urgent,
guaranteed, limited, offer, discount, deal, credit,
loan, money, bank, account, verify, confirm,
click, link, subscribe, unsubscribe, reply, stop,
call, mobile, phone, service, congratulations,
selected, lucky, reward, bonus, opportunity, now

📁 Output Files

1. `balanced_cleaned_dataset_for_ml.csv`

- **Columns:** label, cleaned_message
- **Rows:** 100,000 (50,000 spam + 50,000 ham)
- **Format:** Cleaned, lowercase, lemmatized text
- **Location:** Google Drive & Local